GOOD NOTATION
THE DATA AND AI SCHOOL

DATA ANALYST

INDUSTRY-ORIENTEDCOURSESYLLABUS | 2026 EDITION

From raw data to business decisions — learn the complete analytics workflow.

Advanced Excel • SQL • Python • Statistics • Power BI • Tableau fundamentals • AI-assisted analytics • Portfolio projects

Duration: 24 weeks (approximately 120–150 guided hours)
Level: Beginner to job-ready foundation
Prerequisites: Basic computer knowledge; no prior coding required

Practical learning • Real datasets • Interview preparation • Project portfolio

Curriculum focus: data extraction, data quality, analysis, dashboards, business communication, and responsible use of AI.

1. Course Overview

Thecourse connectstools intoone workflow: understand a business question, collect data, clean it, analyse it, communicate the result, and recommend an action. It is designed for classroom training or structured self-study.

Area	Coverage	Tools
Spreadsheet analysis	Cleaning, formulas, lookups, pivots, dashboards	Excel, Power Query
Database querying	Joins, aggregations, CTEs, window functions, query quality	SQL Server / PostgreSQL / MySQL
Programming	DataFrames, cleaning, EDA, visualisation, automation	Python, Pandas, NumPy
Business intelligence	Data models, DAX, interactive reports	Power BI; Tableau overview
Analytics & AI	Statistics, KPIs, AI-assisted analysis, governance	Excel, SQL, Python, BI, approved AI assistant

Suggested learning rhythm

- 60% hands-on practice with realistic business datasets.
- 20% concepts: KPIs, statistics, data quality, and analytical reasoning.
- 20% communication: explain findings, document assumptions, and present dashboards.

Assessment:weeklylabs,SQL/Pythonquizzes,twomini-projects, dashboard review, mock interviews, and a final capstone.

2. Detailed Module Syllabus

Module1—AnalyticsFoundations&Business Thinking (Week 1 | 4–5 hours)

Outcome: Translate business requests into answerable analytical questions.

- Data analyst role; descriptive, diagnostic, predictive, and prescriptive analytics.
- Business problem framing, stakeholders, assumptions, scope, and success criteria.
- Data types, structured/semi-structured data, files, databases, APIs, and data lifecycle.
- Data quality, privacy, access control, documentation, KPIs, dimensions, and measures.

Practical lab: Write a one-page analysis brief for a sales, customer-support, or operations problem.

Module 2 — Advanced Excel for Analytics (Weeks 2–4 | 12–15 hours)

Outcome: Clean, analyse, summarise, and present spreadsheet data.

- Workbook hygiene, tables, sorting, filtering, validation, and data types.
- IF/IFS, AND/OR, IFERROR, SUMIFS, COUNTIFS, AVERAGEIFS, text/date functions, ROUND.
- XLOOKUP, VLOOKUP, INDEX-MATCH; dynamic arrays such as FILTER, SORT, UNIQUE where supported.
- Cleaning blanks, duplicates, inconsistent labels/dates; TRIM, CLEAN, text-to-columns.
- PivotTables, PivotCharts, slicers, timelines, conditional formatting, dashboard layout.
- Power Query: import, change types, split/merge columns, append, merge, refresh; Power Pivot overview.

Practical lab: Build a monthly sales dashboard with product, region, salesperson, revenue, margin, and target attainment.

Module 3 — SQL & Relational Databases (Weeks 5–8 | 18–22 hours)

Outcome: Query and combine data accurately from relational databases.

- Tables, keys, relationships, schemas, ER diagrams, NULL, and data types.
- SELECT, DISTINCT, aliases, WHERE, AND/OR/NOT, IN, BETWEEN, LIKE, ORDER BY, TOP/LIMIT.
- COUNT, SUM, AVG, MIN, MAX; GROUP BY and HAVING.
- INNER, LEFT, RIGHT, FULL OUTER and self joins; duplicate-row traps.
- CASE, COALESCE/ISNULL, date/string functions, subqueries, CTEs, UNION vs UNION ALL.
- Window functions: ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD, running totals, PARTITION BY.
- Views, temporary tables, transactions, permissions, indexes, and query-plan basics. Practical lab: Analyse orders, customers, products, revenue, repeat purchases, monthly growth, and top-N products.

Module 4 — Statistics & Analytical Reasoning (Weeks 9–10 | 8–10 hours)

Outcome: Interpret data responsibly and distinguish patterns from noise.

- Mean, median, mode, range, variance, standard deviation, quartiles, percentiles, and outliers.
- Distributions, sampling, population vs sample, bias, and probability basics.
- Correlation vs causation, confounding factors, and Simpson's paradox awareness.
- Confidence intervals, hypothesis testing, p-values, practical significance, and A/B testing.
- Regression interpretation, trend, seasonality, and limitations.

Practical lab: Compare two marketing campaigns and write an evidence-based recommendation with caveats.

Module 5 — Python for Data Analytics (Weeks 11–14 | 18–22 hours)

Outcome: Use Python for repeatable analysis and automation.

- Python basics: variables, data types, conditions, loops, functions, exceptions, and Jupyter.
- NumPy arrays, shapes, indexing, vectorised operations, and numerical summaries.
- Pandas Series/DataFrames; read_csv/read_excel, head, info, describe, dtypes, shape.
- Filtering, sorting, loc/iloc, type conversion, string/date operations, missing values, duplicates.
- groupby, agg, value_counts, merge, concat, pivot_table, melt, and reshaping.
- EDA, data validation, distribution checks, and correlation caveats.
- Matplotlib/Seaborn: bar, line, histogram, box, scatter, and heatmap.
- CSV/Excel export, basic SQL/API connections, safe credential handling, and recurring report automation.

Practical lab: Create a notebook that cleans a messy customer dataset and generates an EDA report.

Module 6 — Power BI & Business Intelligence (Weeks 15–18 | 16–20 hours)

Outcome: Build reliable data models and interactive reports.

- Power BI workflow, data sources, import vs DirectQuery concepts, refresh basics.
- Power Query transformations, errors, append/merge, parameters overview.
- Fact/dimension tables, star schema, relationships, cardinality, filter direction, date table.
- DAX: measures vs calculated columns, row/filter context, SUM, COUNTROWS, DISTINCTCOUNT, DIVIDE.
- CALCULATE, FILTER, iterators such as SUMX, and time-intelligence basics.
- Drill-through, tooltips, bookmarks, slicers, conditional formatting, accessibility.
- KPI cards, variance to target, trend analysis, dashboard design, performance, RLS overview.
- AI-assisted report and DAX features where licensed and available; validate generated output.

Practical lab: Create an executive report with a documented data model, KPI definitions, and insight summary.

Module 7 — Tableau & Alternative BI Tools (Week 19 | 4–6 hours)

Outcome: Understand transferable visual analytics concepts and build a basic Tableau dashboard.

- Connect to files/databases; dimensions, measures, discrete and continuous fields.
- Calculated fields, filters, groups, sorting, parameters, and table calculations.
- Worksheets, dashboards, actions, formatting, and stakeholder-friendly design.
- Power BI vs Tableau: shared concepts and tool-selection considerations.

Practical lab: Recreate one Power BI view in Tableau, or document the equivalent design if unavailable.

Module 8 — AI-Assisted Analytics & Modern Data Practices (Week 20 | 4–6 hours)

Outcome: Use AI to accelerate analysis while preserving accuracy, privacy, and accountability.

- Prompt AI for SQL drafts, Python explanations, formulas, chart ideas, and report narratives.

- Use AI to suggest tests, explain errors, propose quality checks, and document workflows.
- Verify outputs by running queries, checking totals, testing edge cases, and comparing known results.
- Hallucinations, confidential data, bias, explainability, intellectual property, and human review.
- Cloud analytics awareness: Microsoft Fabric, lakehouse, semantic models, and pipelines overview.
- Copilot/AI feature availability depends on licence, tenant, and region.

Practical lab: Have AI draft a query or analysis, then submit a verification checklist and corrected result.

Module 9 — Communication, Portfolio & Interview Readiness (Weeks 21–22 | 8–10 hours)

Outcome: Explain findings to technical and non-technical audiences.

- Data storytelling: context, finding, evidence, implication, recommendation, next step.
- Concise insight summaries, dashboard annotations, executive updates, and documentation.
- Requirements gathering, stakeholder communication, assumptions, and KPI definitions.
- Git/GitHub basics, README files, data dictionaries, source notes, and reproducibility.
- Resume project bullets, SQL/Excel/Power BI/Python interview practice, analytical case studies.

Practical lab: Present a five-minute dashboard walkthrough and defend assumptions and data quality decisions.

3. Capstone Projects (Weeks 23–24)

Complete at least two mini-projects and one end-to-end capstone. Document the business question, data dictionary, cleaning decisions, analysis, dashboard, and recommendations.

Project	Deliverables	Skills demonstrated
Retail / E-commerce Sales	SQL analysis, Excel cleaning, Power BI dashboard, five insights	Joins, KPIs, revenue/margin, time trends
Customer Churn & Retention	EDA notebook, segment analysis, churn indicators, recommendations	Pandas, statistics, segmentation
HR / Workforce Analytics	Attrition dashboard, quality log, privacy-aware summary	Data modelling, measures, responsible analysis
Optional: Marketing Campaign	Funnel report and A/B test interpretation	Campaign KPIs, statistics, storytelling

Capstone evaluation rubric

Criteria	Weight	Evidence
Business framing	10%	Clear question, stakeholder, KPI definitions
Data quality	20%	Reproducible cleaning and assumptions
Analysis accuracy	25%	Correct queries/calculations and interpretation
Dashboard design	20%	Readable, useful, accessible, decision-oriented
Communication	15%	Evidence-based actions and clear presentation
Documentation & ethics	10%	README, source notes, privacy, limitations

4. Suggested 24-Week Delivery Plan

Week	Focus
1	Foundations, business questions, KPIs
2	Excel foundations, cleaning, sorting, filtering
3	Advanced formulas, lookups, PivotTables, charts
4	Power Query, dashboard, Excel assessment
5	SQL fundamentals, filtering, sorting
6	Aggregations, GROUP BY, HAVING, CASE
7	Joins, NULL handling, dates, strings
8	CTEs, subqueries, window functions, SQL case study
9	Descriptive statistics, distributions, outliers
10	Sampling, hypothesis tests, A/B testing
11	Python basics, functions, notebooks
12	NumPy and Pandas foundations
13	Pandas cleaning, merge, groupby, reshape
14	EDA, visualisation, automation mini-project
15	Power BI sources and Power Query
16	Data modelling, relationships, star schema
17	DAX measures, CALCULATE, time intelligence
18	Interactive dashboard and review
19	Tableau fundamentals and dashboard practice
20	AI-assisted analytics, verification, governance
21	Business storytelling and portfolio
22	Interview preparation and case studies
23	Capstone: sourcing, cleaning, analysis
24	Capstone dashboard, presentation, mock interview

Suggestedworkload:120–150guidedhoursplusindependentpractice.Adapt to batch schedule and learner pace.

5. Tools & Learning Resources

Recommendedsoftware: Microsoft Excel (Microsoft 365or compatible version), Power BI Desktop, a SQL environment (SQL Server/PostgreSQL/MySQL), Python 3 with Jupyter, Pandas, NumPy, Matplotlib and Seaborn, and Tableau Public or equivalent for introductory practice.

Dataset sources: public government data portals, Kaggle, UCI Machine Learning Repository, or instructor-created synthetic datasets. Check dataset licences and remove personal or confidential information.

Official starting points:

- Microsoft Learn — Data analytics: https://learn.microsoft.com/en-us/training/paths/data-analytics-microsoft/
- Microsoft Learn — Advanced analytics in Power BI: https://learn.microsoft.com/en-us/training/modules/ai-visuals-power-bi/2-visual
- Microsoft — AI data analytics overview: https://www.microsoft.com/en-us/microsoft-copilot/copilot-101/ai-for-data-analysis

These references are starting points. Verify current product requirements and licence access before teaching features such as Copilot.

6. Learning Outcomes & Job-Readiness Checklist

- Translate business requests intomeasurablequestions andwell-definedKPIs.
- Clean spreadsheet data with formulas, PivotTables, and Power Query.
- Write SQL using joins, aggregations, CTEs, and window functions.
- Use Python/Pandas for cleaning, EDA, visualisation, and repeatable analysis.
- Explain basic statistical results and avoid common interpretation errors.
- Build a Power BI model and interactive dashboard with appropriate DAX measures.
- Use AI assistants responsibly and verify generated code, formulas, and insights.
- Present two mini-projects and one documented capstone in a portfolio.
- Explain methods, assumptions, limitations, and recommendations in interviews.

Important note

This is an industry-oriented training curriculum, not a guarantee of employment or a certification syllabus. Hiring requirements vary by company and role. Keep the course current by reviewing job descriptions, tool updates, and the business problems used in practice.

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